LUÍS **ABRANTES**

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SUMMARY

I am currently pursuing a Master's degree in Electrical and Computer Engineering at Instituto Superior Técnico, specializing in Control, Robotics, and AI. With a strong academic background and a passion for innovation and automation, I aim to build a career in aerial robotics, with a strong focus on guidance, navigation, and control for autonomous flight systems.

EDUCATION

IST | Lisbon, Portugal
MSc Electrical and Computer Engineering
Major in Control, Robotics and AI
September 2024 - Present

NOVA FCT | Caparica, Portugal
BSc Electrical and Computer Engineering
September 2021 - July 2024
Grade: 17

SKILLS

Languages
Portuguese (Native)
English (Fluent)

Programming/Softwares
MATLAB
C/C++
Python
Java
Assembly
SQL
ROS
Fusion

Soft Skills
Communication
Teamwork
Problem-Solving
Adaptability
Time Management
Stress Management

EXPERIENCE

Electrical Technical Director | TLMoto, IST
November 2025 - Present

I coordinate the Powertrain, Electronics, and Software departments, ensuring alignment across the overall electrical architecture and development milestones. My role includes driving the electrical system concept definition, managing cross-functional workflows, planning and prioritizing deliverables, and overseeing risk assessment and technical decision-making - *https://tlmoto.tecnico.ulisboa.pt/*

Assistant Researcher for Project Capture | ISR
November 2024 - Present

In this project, I contributed to the development of the gripper design and control mechanism for both the simulation in Gazebo and the real gripper. During this work, I worked with ROS2 (C++ and Python), PX4, and Gazebo Classic, developing Gazebo plugins and ROS2 nodes.
https://capture.isr.tecnico.ulisboa.pt/

Powertrain Member | TLMoto, IST
November 2024 - November 2025 | (1 Year)

As a member of the Powertrain team, I have been responsible for the development of the high-voltage system, including battery pack design, race powertrain simulations, motor bench testing and inverter programming - *https://tlmoto.tecnico.ulisboa.pt/*

Assistant Researcher for Project PackBot | ISR
March 2025 - June 2025 | (4 Months)

During this period, I contributed to the research and testing of computer vision algorithms for 6D object detection and localisation, supporting the integration of cameras into the project development.

Research Internship Project Capture | UNINOVA
January 2024 - February 2024 | (2 Months)

In this project, I assisted in the development of a shuttle drone designed for launch and capture in both cooperative and non-cooperative scenarios - *https://capture.isr.tecnico.ulisboa.pt/*

Member of AEFCT | FCT-UNL
September 2021 - July 2024 | (2 Years, 10 Months)

During this period, I played a key role in developing and expanding the esports department at FCT, contributing significantly to its growth and success.